

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No- 12	Team No- 25	Project Title-Smart Restaurant Ordering & Kitchen Display System
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1. Abstract

Traditional restaurant ordering systems often depend on manual order taking and communication between customers, waiters, and kitchen staff. These methods can lead to incorrect orders, delays, and difficulties in tracking orders and managing restaurant operations. The proposed Smart Restaurant Ordering & Kitchen Display System provides a digital solution to simplify restaurant ordering and kitchen management.

The system allows customers to scan a QR code placed on their table and access the restaurant's digital menu using their mobile device. Customers can browse food items, select quantities, add items to a cart, and place orders directly. The order is stored in the database and displayed on the kitchen dashboard, where kitchen staff can view incoming orders and update their status as Pending, Preparing, Ready, or Delivered. Using Socket.io, order and status updates are communicated in real time.

An admin module is provided to manage menu items, restaurant tables, orders, and basic sales information. The system also calculates bills automatically and provides a printable invoice. The application uses React.js, JavaScript, HTML5, and CSS3 for the frontend, while Node.js and Express.js are used for backend development and REST API implementation. MongoDB is used to store application data such as users, menu items, tables, orders, and order items. QR Code Generation enables table-based ordering, and Git/GitHub is used for version control.

The proposed system aims to reduce manual errors, improve order processing, and provide an efficient and user-friendly digital restaurant management solution.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

1. React.js
2. Node.js
3. Express.js
4. MongoDB
5. Socket.io
6. JavaScript
7. HTML5 & CSS3
8. REST APIs
9. QR Code Generation
10. Git / GitHub
11. Antigravity

Guide Name and Signature_____